

Analog CMOS Experiments on a Reproducible Workbench

MOS device modelling, logic, amplification and oscillation - end-to-end on Crucible

Six self-contained analog CMOS experiments, each combining a Python design-equation check with a SPICE simulation deck - every step driven by the Crucible CLI (igny), in reproducible workspaces, with no manual EDA installation.

#	EXPERIMENT	TOOLS (VIA igny)	RESULT
1-6	MOS modeling → logic → amplifiers → oscillator	ngspice, Python	✓ ALL PASS

OVERVIEW

Toolchain & Contents

Each experiment pairs a compact analytical verification flow with a SPICE deck. The Python script checks the design equations, sizing and operating-point assumptions; the SPICE deck confirms them through the Crucible-managed ngspice tool. Environments and workspaces are created once and bound per experiment. All tool execution runs through igny run.

TOOLCHAIN USED

ROLE	TOOL
Interface	Crucible CLI by Ignition IO
Workbench	Crucible, Community tier
Simulator	Ngspice 46
Analytical checks	Python + Matplotlib

CONTENTS

- 01MOSFET Parameter Extraction
- 02Short-Channel nMOSFET Bias
- 03CMOS & Pseudo-nMOS NAND
- 04Common-Source Amplifier
- 05CMOS Differential Amplifier
- 06Op-Amp Square-Wave Generator

SECTION 1

Crucible Workspace Setup

One environment for the whole lab. Create it once, install the tools once, then bind each experiment to a workspace. Every command below runs through igny.

SETUP COMMAND PATTERN

```
# launch Crucible
igny init

# one-time environment + tools
igny env create analog_experiments
igny workspace create --path analog_demo --env analog_experiments
cd analog_demo
igny tool install ngspice
igny tool install matplotlib

# per experiment (run from its folder)
igny run script .\scripts\run_analysis.py
igny run ngspice -b -o build/spice.log spice/design.cir
type .\build\spice.log
```

How the two checks differ

- The Python script is the analytical check, it verifies the design equations and prints a PASS line, independent of the simulator.
- The SPICE deck is the simulator-backed check. Use -b (batch) for DC operating-point decks; drop -b only when a transient waveform should open in the ngspice GUI (Experiment 6).

EXPERIMENT 01 · DEVICE MODELING

MOSFET Parameter Extraction

AIM

Calculate representative MOSFET parameters - oxide capacitance, mobility-derived process gain, transconductance, output resistance and threshold voltages - for a first-order MOS process.

THEORY

Oxide capacitance per unit area is set by oxide permittivity and thickness. Mobility, oxide capacitance and geometry fix the process-gain terms used in first-order MOS design.

With a chosen overdrive, the flow estimates drain current, transconductance and output resistance, and confirms the parameters are positive and internally consistent.

DESIGN RELATION

$$\begin{aligned}C_{ox} &= \epsilon_{ox} / t_{ox} \\ k_n &= \mu_n \cdot C_{ox} \cdot (W/L) \\ k_p &= \mu_p \cdot C_{ox} \cdot (W/L)\end{aligned}$$

APPARATUS

```
run_analysis.py
design.cir
ngspice
```

PROCEDURE

```
cd experiment_folder
igny workspace create --env analog_experiments
igny run script run_analysis.py
igny run ngspice -b -o build/spice.log spice/design.cir
type .\build\spice.log
```

ANALYTICAL VERIFICATION

```
$ igny run script .\scripts\run_analysis.py
Cox=7.458e-03 F/m^2 kn=3.356e-04 A/V^2
kp=1.193e-04 A/V^2
PASS mosfet_parameter_extraction
```

SPICE RUN

```
$ igny run ngspice -b -o build/spice.log
spice/design.cir
v(dn) = 1.560617e+00
v(dp) = 8.539634e-02
Tool 'ngspice' completed successfully
```

EXPERIMENT 01 · (contd.)

CRUCIBLE EXECUTION

```
C:\Users\ignyt\Desktop\analog_exp >> cd .\mosfet_parameter_extraction\  
C:\Users\ignyt\Desktop\analog_exp\mosfet_parameter_extraction >> igny run script .\scripts\run_analysis.py  
Cox=7.458e-03 F/m^2 kn=3.356e-04 A/V^2 kp=1.193e-04 A/V^2  
PASS mosfet_parameter_extraction  
Script completed successfully  
C:\Users\ignyt\Desktop\analog_exp\mosfet_parameter_extraction >> igny run ngspice -b -o build/spice.log spice/design.cir  
C:\Users\ignyt\Desktop\analog_exp\mosfet_parameter_extraction >> type .\build\spice.log  
  
Note: No compatibility mode selected!  
  
Circuit: MOSFET Parameter Extraction  
  
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000  
  
Using SPARSE 1.3 as Direct Linear Solver  
  
No. of Data Rows : 1  
v(dn) = 1.560617e+00  
v(dp) = 8.539634e-02  
Note: Simulation executed from .control section  
C:\Users\ignyt\Desktop\analog_exp\mosfet_parameter_extraction >> |
```

RESULT

The extraction flow completes cleanly. Oxide capacitance and process-gain values are valid, and the SPICE operating point confirms the NMOS and PMOS models bias correctly.



Experiment 1 - PASS

Key result: $C_{ox}=7.458e-03 \text{ F/m}^2$, $k_n=3.356e-04 \text{ A/V}^2$.

EXPERIMENT 02 · DEVICE BIASING

Properly Biased Short-Channel nMOSFET

AIM

Design a short-channel nMOSFET current sink biased in saturation, and verify width, bias voltages, drain current, transconductance and saturation margin.

THEORY

For a short-channel current sink, VGS must supply enough overdrive while VDS keeps the device in saturation.

The flow picks a target drain current, computes the width for a fixed channel length, and reports the saturation margin so the operating distance from triode is visible.

DESIGN RELATION

$$\begin{aligned}V_{OV} &= V_{GS} - V_{TH} \\ V_{DS} &\geq V_{OV}\end{aligned}$$

APPARATUS

```
run_analysis.py
design.cir
ngspice
```

PROCEDURE

```
cd experiment_folder
igny workspace create --env analog_experiments
igny run script run_analysis.py
igny run ngspice -b -o build/spice.log spice/design.cir
type .\build\spice.log
```

ANALYTICAL VERIFICATION

```
$ igny run script run_analysis.py
W=3.46 um L=0.18 um VGS=0.65 V VDS=0.90 V
ID=100.0 uA gm=1.00 mS margin=0.70 V
PASS short_channel_nmos_bias
```

SPICE RUN

```
$ igny run ngspice -b -o build/spice.log
spice/design.cir
i(vd) = -7.96569e-05
v(d) = 9.000000e-01 v(g) = 6.500000e-01
Tool 'ngspice' completed successfully
```

EXPERIMENT 02 · (contd.)

CRUCIBLE EXECUTION

```
C:\Users\ignyt\Desktop\analog_exp\short_channel_nmos_bias >> igny run script .\scripts\run_analysis.py
W=4.50 um L=0.18 um VGS=0.65 V VDS=0.90 V
ID=100.0 uA gm=1.00 mS saturation_margin=0.70 V
PASS short_channel_nmos_bias
Script completed successfully
C:\Users\ignyt\Desktop\analog_exp\short_channel_nmos_bias >> igny run ngspice -b -o build/spice.log spice/design.cir
C:\Users\ignyt\Desktop\analog_exp\short_channel_nmos_bias >> type .\build\spice.log

Note: No compatibility mode selected!

Circuit: Short Channel NMOS Bias

Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 1
i(vd) = -7.96569e-05
v(d) = 9.000000e-01
v(g) = 6.500000e-01
Note: Simulation executed from .control section
C:\Users\ignyt\Desktop\analog_exp\short_channel_nmos_bias >> |
```

RESULT

The selected width supports the target 100 μ A drain current with a 0.70 V saturation margin - properly biased for the stated operating point.



Experiment 2 - PASS

Key result: ID=100.0 μ A, gm=1.00 mS, margin 0.70 V.

EXPERIMENT 03 · LOGIC FAMILIES

CMOS and Pseudo-nMOS 2-Input NAND Gate

AIM

Design a 2-input NAND in complementary CMOS and pseudo-nMOS, then compare truth-table behaviour and the static low-level output.

THEORY

A complementary CMOS NAND uses a parallel pMOS pull-up and a series nMOS pull-down; the output is low only when both inputs are high.

Pseudo-nMOS replaces the pull-up with an always-on weak pMOS load, cutting device count but making VOL depend on the pull-up / pull-down strength ratio.

DESIGN RELATION

$$Y = \neg(A \cdot B)$$

VOL depends on pMOS-load / nMOS-stack ratio

APPARATUS

run_analysis.py
design.cir
ngspice

PROCEDURE

```
cd experiment_folder
igny workspace create --env analog_experiments
igny run script run_analysis.py
igny run ngspice -b -o build/spice.log spice/design.cir
type .\build\spice.log
```

ANALYTICAL VERIFICATION

```
$ igny run script run_analysis.py
A=0 B=0 CMOS=1 pseudo_nmos=1.800 V
A=1 B=1 CMOS=0 pseudo_nmos=0.212 V
PASS cmos_pseudo_nmos_nand
```

SPICE RUN

```
$ igny run ngspice -b -o build/spice.log
spice/design.cir
v(y1) = 2.413333e-09
v(y2) = 4.067857e-01
Tool 'ngspice' completed successfully
```

EXPERIMENT 03 · (contd.)

CRUCIBLE EXECUTION

```
C:\Users\ignyt\Desktop\analog_exp >> cd .\cmos_pseudo_nmos_nand\  
C:\Users\ignyt\Desktop\analog_exp\cmos_pseudo_nmos_nand >> igny run script .\scripts\run_analysis.py  
A=0 B=0 CMOS=1 pseudo_nmos=1.800 V  
A=0 B=1 CMOS=1 pseudo_nmos=1.800 V  
A=1 B=0 CMOS=1 pseudo_nmos=1.800 V  
A=1 B=1 CMOS=0 pseudo_nmos=0.212 V  
PASS cmos_pseudo_nmos_nand  
Script completed successfully  
C:\Users\ignyt\Desktop\analog_exp\cmos_pseudo_nmos_nand >> igny run ngspice -b -o build/spice.log spice/design.cir  
C:\Users\ignyt\Desktop\analog_exp\cmos_pseudo_nmos_nand >> type .\build\spice.log  
  
Note: No compatibility mode selected!  
  
Circuit: CMOS and Pseudo NMOS NAND  
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000  
Using SPARSE 1.3 as Direct Linear Solver  
No. of Data Rows : 1  
v(y1) = 2.413333e-09  
v(y2) = 4.067857e-01  
Note: Simulation executed from .control section  
C:\Users\ignyt\Desktop\analog_exp\cmos_pseudo_nmos_nand >> |
```

RESULT

Both styles implement NAND. Pseudo-nMOS holds 1.800 V for pull-up cases and reaches 0.212 V when both inputs are high.



Experiment 3 - PASS

Key result: Pseudo-nMOS low output 0.212 V.

EXPERIMENT 04 · ANALOG GAIN STAGE

Common-Source Amplifier

AIM

Design a resistively loaded nMOS common-source amplifier and verify its bias point, small-signal voltage gain, output headroom and dominant-pole estimate.

THEORY

The common-source stage converts input gate-voltage variation into an inverted drain-voltage swing.

The output quiescent voltage must stay inside a useful swing, while drain-node capacitance and output resistance set a dominant-pole estimate.

DESIGN RELATION

$$A_v = -g_m \cdot (R_D \parallel r_o)$$

APPARATUS

```
run_analysis.py
design.cir
ngspice
```

PROCEDURE

```
cd experiment_folder
igny workspace create --env analog_experiments
igny run script run_analysis.py
igny run ngspice -b -o build/spice.log spice/design.cir
type .\build\spice.log
```

ANALYTICAL VERIFICATION

```
$ igny run script run_analysis.py
gm=1.00 mS ro=125.0 kOhm gain=-6.90 V/V
VoutQ=1.05 V pole=230.8 MHz
PASS common_source_amplifier
```

SPICE RUN

```
$ igny run ngspice -b -o build/spice.log
spice/design.cir
v(out) = 1.300673e+00
Tool 'ngspice' completed successfully
```

EXPERIMENT 04 · (contd.)

CRUCIBLE EXECUTION

```

C:\Users\ignyt\Desktop\analog_exp >> cd .\common_source_amplifier\
C:\Users\ignyt\Desktop\analog_exp\common_source_amplifier >> igny run script .\scripts\run_analysis.py
gm=1.00 mS ro=125.0 kOhm gain=-6.90 V/V VoutQ=1.05 V pole=230.8 MHz
PASS common_source_amplifier
Plot saved to C:\Users\ignyt\Desktop\analog_exp\common_source_amplifier\build\common_source_waveform.png
Script completed successfully
C:\Users\ignyt\Desktop\analog_exp\common_source_amplifier >> igny run ngspice -b -o build/spice.log spice/design.cir
C:\Users\ignyt\Desktop\analog_exp\common_source_amplifier >> type .\build\spice.log

```

Note: No compatibility mode selected!

Circuit: Common Source Amplifier

Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

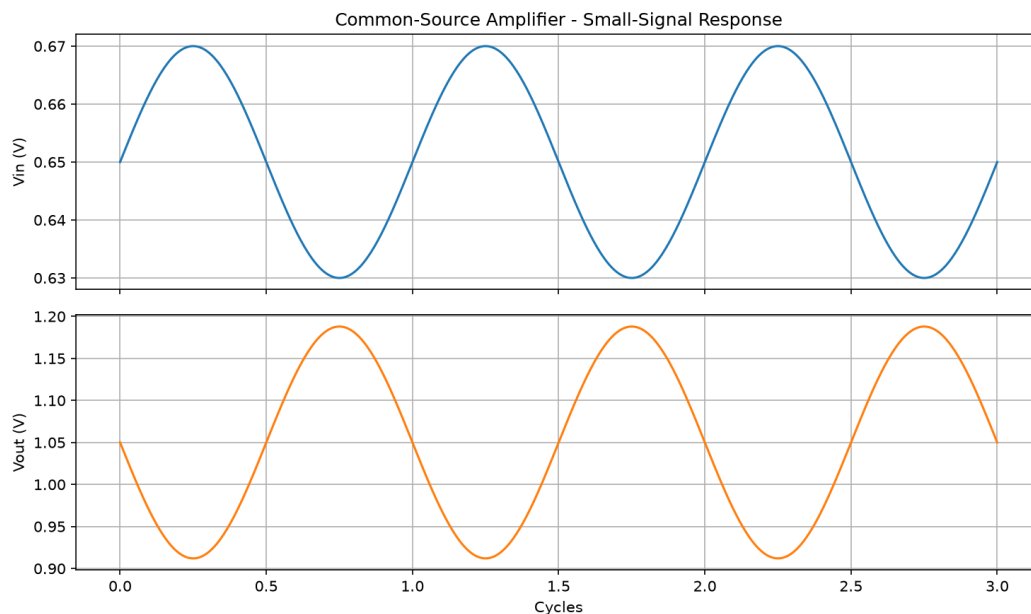
No. of Data Rows : 1

v(out) = 1.300673e+00

Note: Simulation executed from .control section

```
C:\Users\ignyt\Desktop\analog_exp\common_source_amplifier >> |
```

PLOT GENERATED ON CRUCIBLE



RESULT

The amplifier reaches -6.90 V/V gain with a 1.05 V output quiescent point and an estimated 230.8 MHz dominant pole. `plot()` renders the inverted small-signal waveform.

**Experiment 4 - PASS**

Key result: Gain -6.90 V/V, pole 230.8 MHz.

EXPERIMENT 05 · DIFFERENTIAL ANALOG STAGE

CMOS Differential Amplifier

AIM

Design a CMOS differential amplifier and estimate differential gain, common-mode gain, CMRR and input common-mode range.

THEORY

A differential amplifier responds to input difference while rejecting common-mode motion. Key checks are differential gain, tail biasing, common-mode range and CMRR.

The flow computes differential and common-mode gain, reports CMRR in dB, and verifies headroom against the bias assumptions.

DESIGN RELATION

$$\text{CMRR} = |A_d / A_{cm}|$$

APPARATUS

```
run_analysis.py
design.cir
ngspice
```

PROCEDURE

```
cd experiment_folder
igny workspace create --env analog_experiments
igny run script run_analysis.py
igny run ngspice -b -o build/spice.log spice/design.cir
type .\build\spice.log
```

ANALYTICAL VERIFICATION

```
$ igny run script run_analysis.py
Ad=-5.73 V/V Acm=-0.0057 CMRR=60.0 dB
VCM=[0.85,1.65] V
PASS cmos_differential_amplifier
```

SPICE RUN

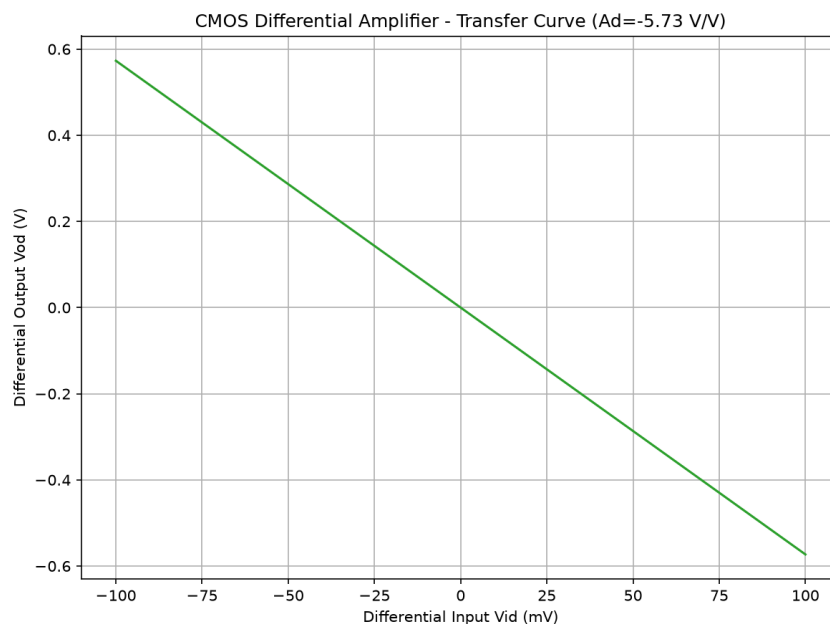
```
$ igny run ngspice -b -o build/spice.log
spice/design.cir
v(outp) = 2.691222e-01
v(outn) = 1.119067e+00
Tool 'ngspice' completed successfully
```

EXPERIMENT 05 · (contd.)

CRUCIBLE EXECUTION

```
C:\Users\ignyt\Desktop\analog_exp >> cd .\cmos_differential_amplifier\  
C:\Users\ignyt\Desktop\analog_exp\cmos_differential_amplifier >> igny run script .\scripts\run_analysis.py  
Ad=-5.73 V/V Acm=-0.0057 CMRR=60.0 dB VCM=[0.85,1.65] V  
PASS cmos_differential_amplifier  
Plot saved to C:\Users\ignyt\Desktop\analog_exp\cmos_differential_amplifier\build\differential_transfer.png  
Script completed successfully  
C:\Users\ignyt\Desktop\analog_exp\cmos_differential_amplifier >> igny run ngspice -b -o build/spice.log spice/design.cir  
C:\Users\ignyt\Desktop\analog_exp\cmos_differential_amplifier >> type .\build\spice.log  
  
Note: No compatibility mode selected!  
  
Circuit: CMOS Differential Amplifier  
  
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000  
  
Using SPARSE 1.3 as Direct Linear Solver  
  
No. of Data Rows : 1  
v(outp) = 2.691222e-01  
v(outn) = 1.119067e+00  
Note: Simulation executed from .control section  
C:\Users\ignyt\Desktop\analog_exp\cmos_differential_amplifier >> |
```

PLOT GENERATED ON CRUCIBLE



RESULT

The amplifier reaches -5.73 V/V differential gain with 60.0 dB CMRR and a valid 0.85–1.65 V input common-mode range. plot() renders the Vod–Vid transfer curve.



Experiment 5 - PASS

Key result: Gain -5.73 V/V, CMRR 60.0 dB.

EXPERIMENT 06 · OSCILLATOR

CMOS Op-Amp Square-Wave Generator

AIM

Design an op-amp relaxation oscillator that produces a square wave above 1 MHz, and verify it in an interactive ngspice transient run.

THEORY

The circuit is a Schmitt-trigger relaxation oscillator: a capacitor charges and discharges between thresholds set by the feedback factor around a mid-supply reference.

The chosen R and C give a frequency above 1 MHz at a near-50% duty cycle; the clamped comparator holds the output between the rails so it oscillates instead of diverging.

DESIGN RELATION

$$f = 1 / (2 \cdot R \cdot C \cdot \ln((1+\beta)/(1-\beta)))$$

APPARATUS

```
run_analysis.py
design.cir
ngspice
```

PROCEDURE

```
cd experiment_folder
igny workspace create --env analog_experiments
igny run script run_analysis.py
igny run ngspice -o build/spice.log spice/design.cir
```

ANALYTICAL VERIFICATION

```
$ igny run script run_analysis.py
R=8.20 kOhm C=50.0 pF beta=0.50
freq=1.110 MHz duty=0.50
PASS opamp_square_wave_generator
```

SPICE RUN

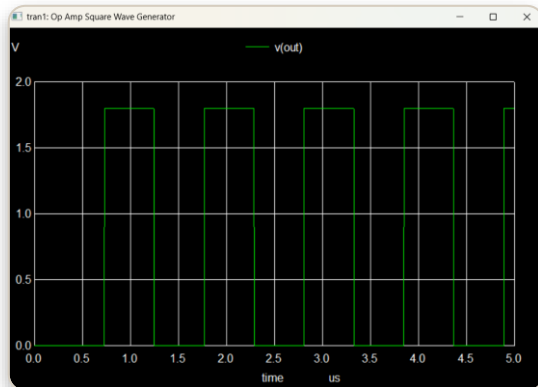
```
$ igny run ngspice -o build/spice.log
spice/design.cir
Transient run - .control opens plot v(out)
Sustained square wave, 0 <-> 1.8 V
```

EXPERIMENT 06 · (contd.)

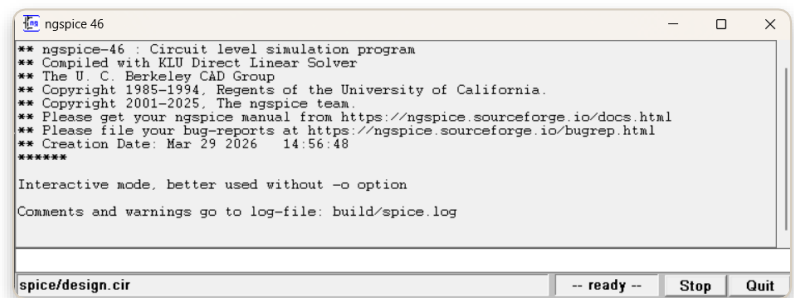
CRUCIBLE EXECUTION

```
C:\Users\ignyt\Desktop\analog_exp >> cd .\opamp_square_wave_generator\  
C:\Users\ignyt\Desktop\analog_exp\opamp_square_wave_generator >> igny run script .\scripts\run_analysis.py  
R=8.20 kOhm C=50.0 pF beta=0.50 freq=1.110 MHz duty=0.50  
PASS opamp_square_wave_generator  
Script completed successfully  
C:\Users\ignyt\Desktop\analog_exp\opamp_square_wave_generator >> igny run ngspice -o build/spice.log spice/design.cir
```

SQUARE WAVE OUTPUT



NGSPICE GUI



RESULT

The RC network gives a 1.110 MHz oscillation at ~50% duty. Run without -b so the .control / plot block opens the live square-wave waveform in the ngspice GUI.



Experiment 6 - PASS

Key result: Frequency 1.110 MHz, duty 0.50.

SUMMARY

Summary of Results

#	EXPERIMENT	KEY RESULT	STATUS
1	MOSFET Parameter Extraction	$C_{ox}=7.458e-03 \text{ F/m}^2$, $k_n=3.356e-04 \text{ A/V}^2$	✓ PASS
2	Properly Biased Short-Channel nMOSFET	$I_D=100.0 \mu\text{A}$, $g_m=1.00 \text{ mS}$, margin 0.70 V	✓ PASS
3	CMOS and Pseudo-nMOS 2-Input NAND Gate	Pseudo-nMOS low output 0.212 V	✓ PASS
4	Common-Source Amplifier	Gain -6.90 V/V , pole 230.8 MHz	✓ PASS
5	CMOS Differential Amplifier	Gain -5.73 V/V , CMRR 60.0 dB	✓ PASS
6	CMOS Op-Amp Square-Wave Generator	Frequency 1.110 MHz, duty 0.50	✓ PASS

All six analog CMOS experiments run successfully through the Crucible CLI. Each pairs design-equation verification with SPICE simulation, tied to a named environment so any engineer can reproduce the lab with identical tool versions - design collateral, simulator execution and environment state kept together in one reproducible workflow.